Fourier Transform Equations

Matthew Davis
m.davis@student.unsw.edu.au

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Discrete Time Fourier Transform (DTFT)

- From discrete time to continuous frequency
- Equation 3.10 in the textbook.

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\theta n} \tag{1}$$

Inverse Discrete Time Fourier Transform

- From continuous frequency to discrete time .
- Equation 3.14 in the textbook.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{j\theta n} d\theta \tag{2}$$

Discrete Fourier Transform (DFT)

- From discrete time to discrete frequency .
- This is the same as the DTFT (equation 1), but we've subbed in $\theta = \frac{2\pi k}{N}$
- This equations samples $X(z)$ at N equally spaced points around the unit circle (starting from $z = 1$).

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\frac{2\pi k}{N}} \text{ for } k = 0, 1, 2, 3 \ldots (N-1)$$

Inverse Discrete Fourier Transform

- From discrete frequency back to discrete time.
- This is similar to the IDTFT (equation 2), but we've but we've subbed in $\theta = \frac{2\pi k}{N}$.

$$x\left[n\right] = \frac{1}{N}\sum_{k=0}^{N-1} X\left[k\right] e^{jn\frac{2\pi}{N}k} \text{ for } n = 0, 1, 2 \ldots (N-1)$$

Continuous Time Fourier Transform (CTFT)

- From continuous time to continuous frequency.

$$F\left(\omega\right) = \int_{-\infty}^{\infty} f\left(t\right) e^{-j\omega t} dt$$

Inverse Continuous Time Fourier Transform (ICTFT)

- From continuous frequency to continuous time.

$$f\left(t\right) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\left(\omega\right) e^{j\omega t} d\omega$$